

Pre-processing and Feature Extraction Pipeline for Hybrid DSP + Neural Noise Suppression

1 Introduction

This document describes the pre-processing and feature extraction pipeline used in a hybrid DSP and neural network-based speech denoising system. The design is based on principles similar to the RNNoise framework, where classical signal processing is used to extract compact and perceptually meaningful features for efficient neural network inference.

2 Signal Pre-processing

2.1 Sampling Configuration

The system operates primarily at a sampling rate of:

- **16 kHz (default)**

An optional high-quality mode supports:

- **48 kHz (experimental)**

2.2 Framing and Windowing

The input audio signal is segmented into short-time frames:

- Frame duration: **20 ms**
- Frame size: **320 samples at 16 kHz**
- Overlap: **50%**
- Window function: **Hann window**

This enables time-localized spectral analysis while ensuring smooth reconstruction.

2.3 Short-Time Fourier Transform (STFT)

Each frame is transformed into the frequency domain using the Fast Fourier Transform (FFT):

- Produces **complex spectrum** (magnitude + phase)

Important:

- Magnitude is used for feature extraction and modification
- Phase is preserved for reconstruction

3 Feature Extraction

Feature extraction is divided into three parallel paths: spectral envelope, pitch, and non-stationarity.

3.1 Spectral Envelope Features

3.1.1 Bark Band Energy Computation

The magnitude spectrum is grouped into perceptual frequency bands:

- **22 Bark bands (default)**
- **24 Bark bands (optional)**

This reduces dimensionality while preserving perceptual relevance.

3.1.2 Log Compression

Logarithmic scaling is applied:

$$E_{log} = \log_{10}(E + \epsilon) \tag{1}$$

This compresses dynamic range and stabilizes learning.

3.1.3 Discrete Cosine Transform (DCT)

A DCT is applied to decorrelate features:

- Produces BFCC-like coefficients
- Improves neural network efficiency

3.1.4 Delta Features

Temporal derivatives are computed:

- Captures spectral dynamics
- Enhances temporal modeling

3.2 Pitch Features

Pitch-related features are extracted using time-domain analysis:

- Pitch period estimation via autocorrelation
- Normalized pitch correlation

From this, multiple features are derived:

- Pitch lag
- Correlation strength
- Harmonic structure indicators

These features help distinguish voiced speech from noise.

3.3 Non-Stationarity Feature

A feature representing spectral variation across frames is computed:

- Based on spectral flux
- Indicates signal stationarity

This helps the model differentiate between:

- Speech (non-stationary)
- Noise (often stationary)

4 Feature Vector Construction

All extracted features are concatenated into a single vector:

- Spectral features (BFCC + delta)
- Pitch features
- Non-stationarity feature

Typical size: ~ 42 features per frame

5 Notes on Design Choices

- Bark bands approximate human auditory perception
- Log + DCT reduces redundancy
- Pitch features improve speech preservation
- Phase is not modified in this stage

6 Experimental Extensions

The following modifications are considered optional and are not part of the baseline system:

- Increasing Bark bands ($22 \rightarrow 24$)
- Removing or replacing DCT with learned transformation
- Adding additional spectral features (centroid, rolloff)
- Using alternative targets such as Ideal Ratio Mask (IRM)

7 Conclusion

This pipeline provides a computationally efficient and perceptually aligned feature representation for neural noise suppression systems, balancing DSP interpretability with neural network flexibility.